PROJECT TITAN

Prototype Development Plan

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EDUCATIONAL USE NOTICE

Project TITAN is an educational engineering design project. It is not a clinical device, regulatory submission, or product approved for human use.

Document Control

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A	July 2026	Initial phased prototype-development plan.	Michael Lozada-Longog

1. Objective

Define a low-risk progression from digital design to a measurable physical prototype while preventing unsupported wearable or clinical claims.

2. Prototype Stages

Stage	Purpose	Build	Exit Criteria
P0 - Digital baseline	Confirm CAD, documentation, and static analysis	Fusion assembly and controlled document set	V1 documentation released; open issues identified
P1 - Fit-check prototype	Validate geometry, clearances, and assembly sequence	FDM printed structural parts plus low-cost shaft/bearing substitutes	Assembles without interference; motion observed; change list captured
P2 - Bench mechanical prototype	Validate real bearings, shaft alignment, and structural assembly	Machined aluminum or hybrid metal/polymer parts; commercial hardware	Dimensional inspection passed; smooth movement; safe bench load test completed
P3 - Instrumented bench concept	Demonstrate sensing only, not clinical function	IMU/encoder/load sensor and data logger	Repeatable angle/load data on a fixture
P4 - Future biomedical research concept	Explore controlled human-factors questions only after approvals	Redesigned system with safety controls	Separate review, risk assessment, and institutional approval required

3. P1 Build Plan

1. Export current components as STL with consistent orientation.
2. Print plates/frames using a strong, dimensionally stable material and adequate wall count.
3. Use purchased bearings only if the printed bore fit is controlled; otherwise use printed bushings for fit checks.
4. Record every interference, gap, and difficult assembly step.
5. Update CAD and drawings before proceeding to metal.

4. P2 Bench Test Plan

Test	Method	Preliminary acceptance
Dimensional inspection	Caliper/micrometer/bore gauges	Critical dimensions within released tolerances
Rotation/kinematics	Manual movement through intended range	No binding, scraping, or uncontrolled looseness
Static deflection	Fixture assembly; apply	No permanent deformation;

	incremental load below test limit	measured deflection recorded
Fastener retention	Torque and witness-mark review	No loosening during bench cycling
Post-test inspection	Visual and dimensional comparison	No cracks, yielding, or bearing migration

5. Safety Boundaries

- No human wear or clinical evaluation in P0-P2.
- Use guards and remote loading during bench tests.
- Define proof-load limits before testing.
- Stop testing upon visible damage, unusual noise, or unexpected displacement.
- Document deviations and failed tests; do not hide them.

6. Required Future Artifacts

Prototype BOM, build traveler, inspection sheet, test protocol, raw data, photographs, deviation log, and test report.